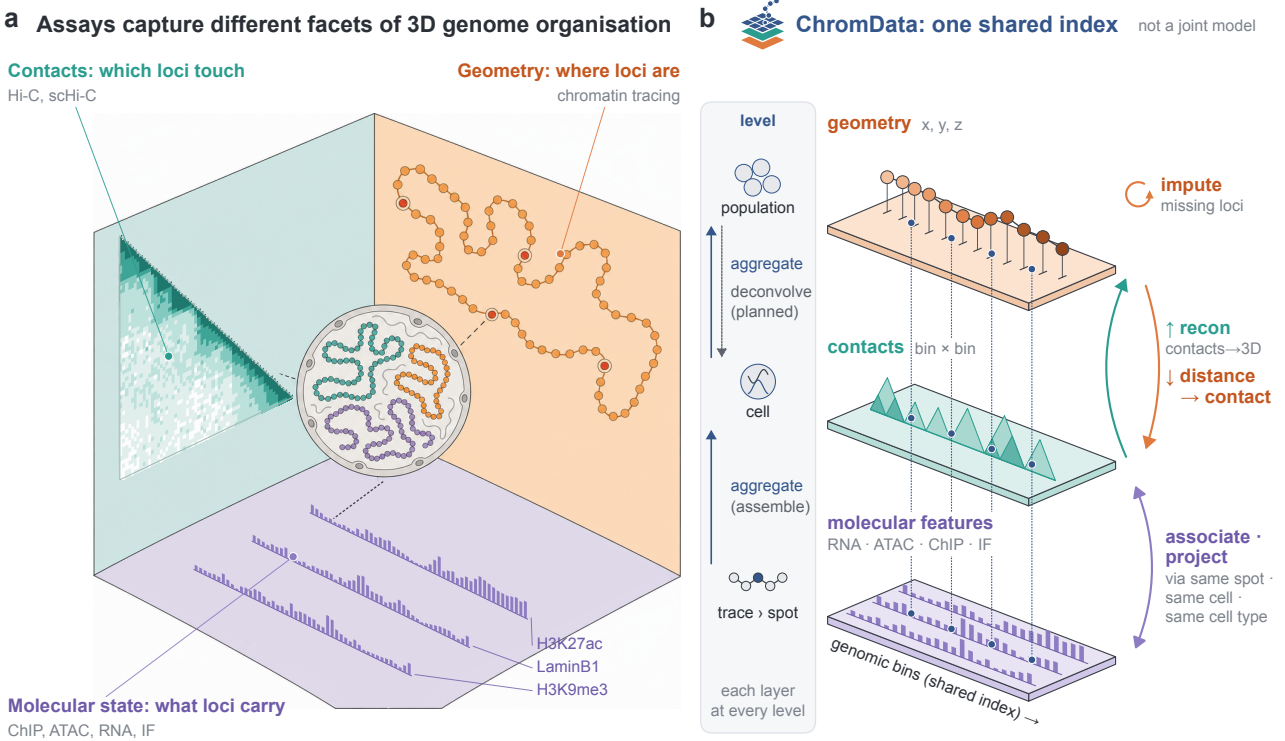


c An open ecosystem

interoperates with 4DN · scverse · Open2C · Juicer

Interfaces
people · agents

python™ API · CLI · web browser · MCP server + agent skills

Methods
one API

- **uchrom.recon** — 3D structures from contacts single-cell and bulk
- **uchrom.im** — spots → traces fill in missing loci
- **uchrom.strc** — TADs, loops, compartments
- **uchrom.fea** — per-locus geometric features
- **uchrom.emb** — cell embeddings across modalities

ChromData cell › trace › spot · geometry · contacts · features · .chromdata.zarr, versioned spec

Formats
& standards

read ↑

- 4DN: FOF-CT
- Juicer: .hic
- Open2C: .pairs, cooler, .cool · .mcool · .scool
- tracing tools: pyHiM, seqFISH+, Bintu et al.
- structures: .3dg

write / link ↓

linked, not copied

- Zarr, Parquet — U-Chrom: .chromdata.zarr, .cdz
- scverse: AnnData, MuData
- Open2C: .mcool, .scool
- structures (export): PDB → PyMOL / ChimeraX